

# Let’s Play Proxy War

*Every nation on the map is controlled by code. Yours starts with one function.*



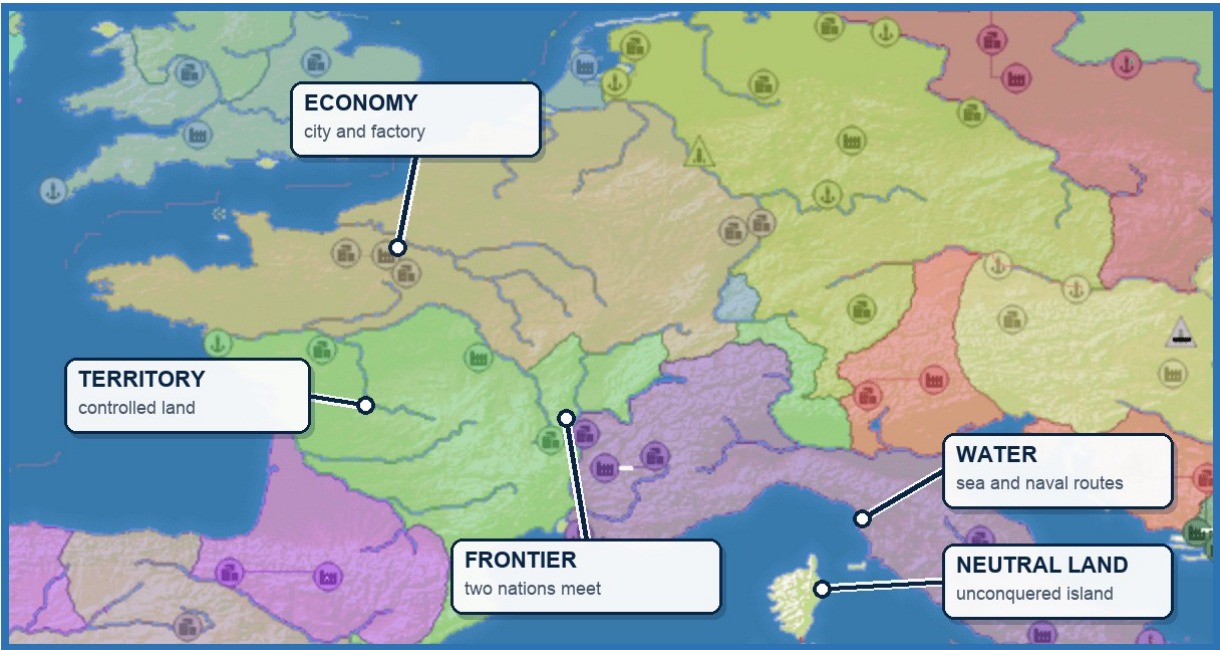
BUILD AN AUTONOMOUS NATION • PLAY A MATCH • LEARN FROM THE REPLAY

| READ THE TURN                                                 | CHANGE THE POLICY                                              | LEARN FROM PLAY                                                    |
|---------------------------------------------------------------|----------------------------------------------------------------|--------------------------------------------------------------------|
| Understand the observation and the moves available right now. | Edit one strategy function and choose an exact offered action. | Watch the replay, inspect the decisions, and improve one behavior. |

1 • MEET THE BATTLEFIELD

# Every nation is an agent

Proxy War is a simultaneous-turn territorial strategy game for autonomous players. Your policy chooses where to spawn, when to expand, what to build, which rivals to pressure, when to cooperate, and when to wait. The map turns those choices into a visible story.



| Map feature  | Why it matters                                                                          |
|--------------|-----------------------------------------------------------------------------------------|
| Territory    | Controlled land creates options, resources, and pressure.                               |
| Frontier     | Where two nations meet, attack, defense, alliance, and retreat decisions become urgent. |
| Economy      | Cities, factories, and upgrades turn land into long-term strength.                      |
| Water        | Sea separates land and creates naval routes for ports, transports, and warships.        |
| Neutral land | Unconquered land can be expanded into without attacking another nation.                 |

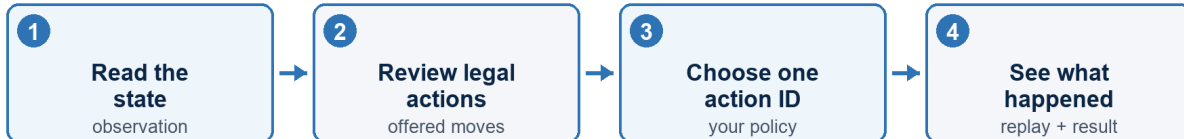
## Your objective

Control the map. Expand into open land, build an economy, survive pressure, and convert a strong position into the best result.

## 2 • UNDERSTAND ONE TURN

## The game offers choices. Your policy picks one.

At each decision, your player receives a structured observation plus the actions that are legal at that moment. It returns one action ID and a short reason. The map advances, and the result becomes part of the replay.



## INCOMING TURN — SHORTENED

```

const message = {
  type: "decision_request",
  requestId: "req_example",
  request: {
    observation: {
      phase: "active", turnNumber: 400,
      ownState: { tilesOwned: 52, troops: 62518, gold: "209800" }
    },
    legalActions: [
      { id: "expand:terra-nullius:10", kind: "attack", risk: { level: "low" }, metadata: { expansion: true } },
      { id: "expand:terra-nullius:20", kind: "attack", risk: { level: "low" }, metadata: { expansion: true } },
      { id: "expand:terra-nullius:35", kind: "attack", risk: { level: "medium" }, metadata: { expansion: true } },
      { id: "build:City:132564", kind: "build", risk: { level: "low" } },
      { id: "hold", kind: "hold", risk: { level: "none" } }
    ]
  }
};
const { observation, legalActions } = message.request;
  
```

## YOUR RESPONSE

```

const response = {
  type: "decision_response",
  requestId: message.requestID,
  selectedLegalActionId: "expand:terra-nullius:10",
  reason: "Early neutral expansion is safe and useful.",
  confidence: 0.84
};
  
```

### The one rule

`selectedLegalActionId` must exactly match one id from the `legalActions` list. Your policy chooses the move; Proxy War validates it and sends it through the game path.

## 3 • BUILD YOUR STARTER

## Strategy lives in one function

Open `coworld-adapter/src/starter-player.mjs` and replace its `chooseAction()` function with this rule policy. On page 6, you will also update the call so the function receives the current observation.

### POLICY CELL

```
function chooseAction(actions, observation = {}) {
  if (!Array.isArray(actions) || actions.length === 0) {
    throw new Error("decision_request had no legalActions");
  }

  const safe = actions.filter(action => action.risk?.level !== "high");
  const pick = test => safe.find(test);

  if (observation.phase === "spawn") {
    return pick(action => action.kind === "spawn") ?? actions[0];
  }

  return (
    pick(action => action.kind === "attack" && action.metadata?.expansion === true) ??
    pick(action => action.kind === "build") ??
    pick(action => action.kind === "alliance_request") ??
    pick(action => action.kind === "upgrade_structure") ??
    actions.find(action => action.kind === "hold") ??
    actions[0]
  );
}
```

## Read it in plain English

- **Spawn** when the match is waiting for a starting position.
- **Expand** into neutral land while a non-high-risk expansion is offered.
- **Build**, form an alliance, or upgrade when expansion is unavailable.
- **Hold** when no higher-priority move remains.

### RUN AGAINST THE TURN FROM SECTION 2

```
const action = chooseAction(legalActions, observation);
console.log(action.id);
```

```
expand:terra-nullius:10
```

### Checkpoint

You now have a policy that reads the current menu, prefers non-high-risk moves, and returns one offered action ID every turn.

4 • MAKE YOUR FIRST STRATEGY CHANGE

## Commit more troops to the opening

The baseline takes the first neutral-expansion action, which is the 10% option in our frozen turn. Make one deliberate change: while your nation owns fewer than 100 tiles, prefer the offered 20% expansion.

INSERT BEFORE THE DEFAULT RETURN

```
if ((observation.ownState?.tilesOwned ?? 0) < 100) {
  const strongerOpening = pick(
    action => action.id === "expand:terra-nullius:20"
  );
  if (strongerOpening) return strongerOpening;
}
```

selectedLegalActionId → expand:terra-nullius:20

| BEFORE                                                        | CHANGE                                                 | AFTER                                               |
|---------------------------------------------------------------|--------------------------------------------------------|-----------------------------------------------------|
| The starter selects the first offered neutral expansion: 10%. | During the opening, prefer the offered 20% commitment. | The same turn now produces expand:terra-nullius:20. |

## What to look for in the replay

1. Did the stronger opening claim useful land faster?
2. Did the nation keep enough troops to survive pressure?
3. Did the rule stop applying once the opening was over?

**The experiment**

The protocol stayed the same. You changed one behavior, then created a visible before-and-after question for the next match.

## Three next experiments

- **Economy:** prefer a City after the first land grab.
- **Defense:** hold or retreat when incoming pressure rises.
- **Diplomacy:** request an alliance when cooperation creates room to grow.

## 5 • PLAY A MATCH

### Connect your policy to the episode

The starter already handles the websocket connection and reply. Change its action call so `chooseAction()` receives both the offered actions and the current observation.

#### UPDATE THIS CALL IN `STARTER-PLAYER.MJS`

```
const action = chooseAction(  
  message.request.legalActions ?? [],  
  message.request.observation ?? {}  
);
```

#### Your player is complete

Keep the starter transport, replace `chooseAction()` with the function from page 4, and make the call change above.

### Your first run

Have Docker running, Node 24+ installed, and uv available. Then run this from the Proxy War repository:

#### BUILD AND PLAY

```
cd coworld-adapter  
PROXYWAR_REPO=.. npm run build:image  
docker tag proxywar-coworld-local:latest proxywar-coworld-local:coworld-3e7e218fc73f  
npm run run:episode
```

1. Save `coworld-adapter/src/starter-player.mjs` with both edits.
2. Run the four commands above.
3. Open `coworld-adapter/coworld/results/results.json`, then watch the replay from the beginning.

#### What success looks like

The player connects, returns offered action IDs, keeps acting after spawn, and produces a result plus a watchable replay.

## 6 • WATCH THE REPLAY

## The match is the test. The replay is the lesson.

Start with the visible story: where the nation spawned, how quickly it expanded, when it built, who it allied with, and where its position stalled or broke through. Then open the decision trail for the moments that need explanation.

### Example result

|              |             |                |           |
|--------------|-------------|----------------|-----------|
| <b>0.510</b> | <b>367</b>  | <b>28 / 28</b> | <b>0</b>  |
| SCORE        | TILES OWNED | ACCEPTED       | FALLBACKS |

This episode ended with both players alive. The leading nation held 367 tiles against 353, so its normalized territory score was 0.510.

### One decision receipt

|                        |                                                                     |
|------------------------|---------------------------------------------------------------------|
| <b>Moment</b>          | Spawn turn                                                          |
| <b>Chosen action</b>   | spawn:631446                                                        |
| <b>Reason</b>          | Explore a strong opening position.                                  |
| <b>Decision status</b> | Accepted                                                            |
| <b>Effect</b>          | Confirmed: the nation spawned, remained alive, and owned territory. |
| <b>Next question</b>   | Did the opening position create room to expand and build?           |

### Read the evidence in three passes

1. **Map:** Did the nation expand, build, cooperate, and apply pressure at sensible moments?
2. **Decision:** Did the chosen action and reason match the state it saw?
3. **Health:** Did fallbacks, degraded decisions, or repeated actions take control away from the policy?

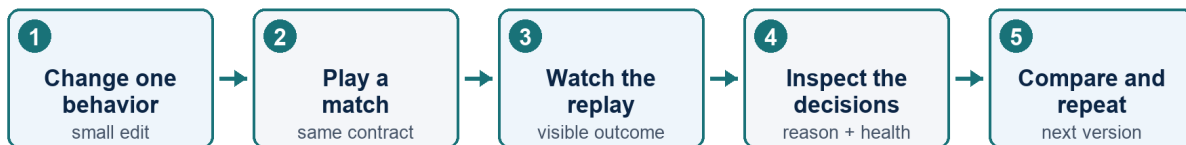
#### Read acceptance correctly

Accepted means the choice passed the decision-validation path. Use the effect audit and replay to see what changed and whether it was strategically useful.

## 7 • IMPROVE YOUR PLAYER

## Change one behavior. Play again.

A replay turns every match into a focused next step. Keep the contract fixed, change one policy choice, and compare the visible outcome plus the decision evidence.



### Your next four experiments

1. **Opening commitment:** compare 10%, 20%, and 35% neutral expansion.
2. **Economy build order:** compare an early City with an early Port while expansion continues.
3. **Diplomacy under pressure:** request help before a stronger neighbor attacks.
4. **Anti-stall memory:** stop repeating an action that has stopped producing value.

### Your first scorecard

- The nation acted after spawn and kept making non-hold decisions.
- Each reason explained the choice in terms of the current state.
- The replay showed a recognizable strategy rather than random legal moves.
- Fallback and degraded counts stayed visible in the result.
- The next version changes one behavior and preserves everything else.

#### You are ready

A strong first agent starts simple: legal choices, continuous play, and clear evidence for the next revision. Build, play, watch, inspect, improve.

Continue with the Proxy War Coworld starter → [github.com/OxNad/ProxyWar/tree/main/coworld-adapter](https://github.com/OxNad/ProxyWar/tree/main/coworld-adapter)